Prescription-time design of Pulseq sequences on GE

Vendor-Agnostic Pulse Sequence Programming with Pulseq II – 26 Feb 2026

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Motivation and goal

Vendor-specific Sequence development environments:

✗ High complexity, cannot be moved to other platforms

✓ Enable flexible prescription time sequence modifications (FOV, matrix, etc)

Vendor-agnostic Sequence development environments:

✓ Easy to learn, fast prototyping, portability

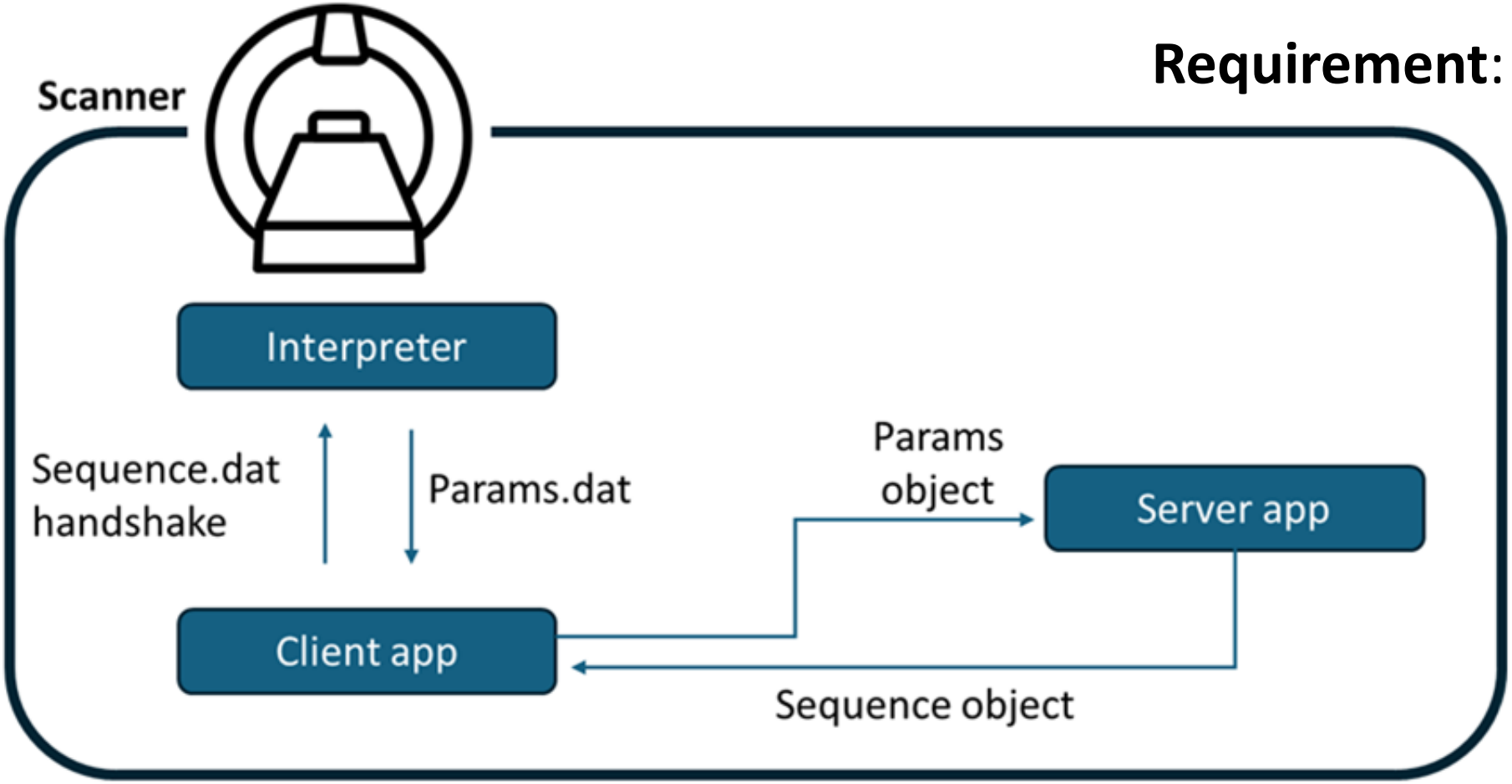
✗ Limited prescription-time customizability

Goal: enable prescription time sequence design using a vendor-agnostic framework

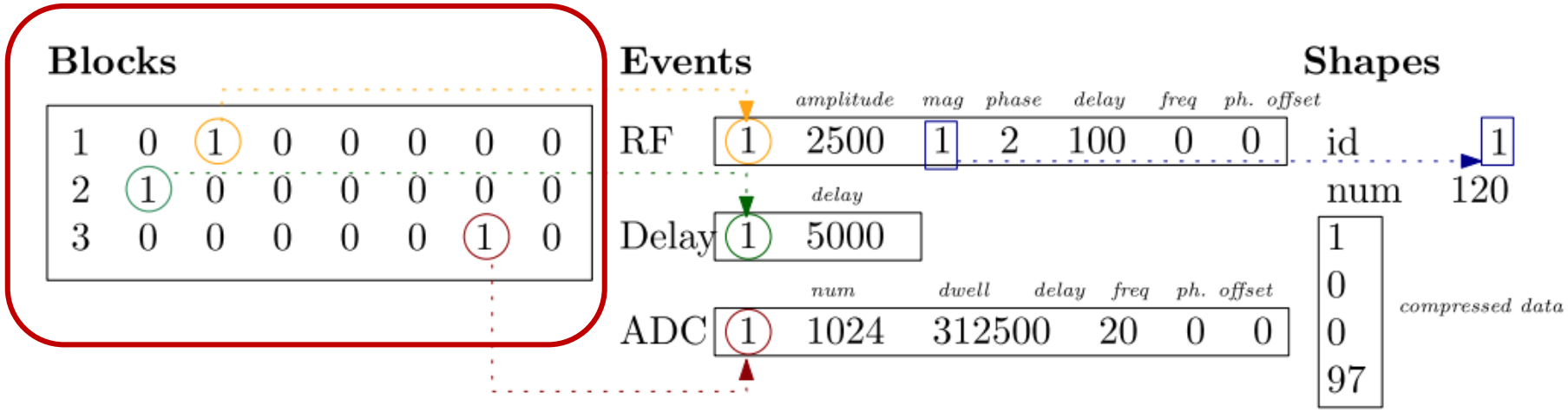
Pulseq

Framework architecture

Requirement: low latency

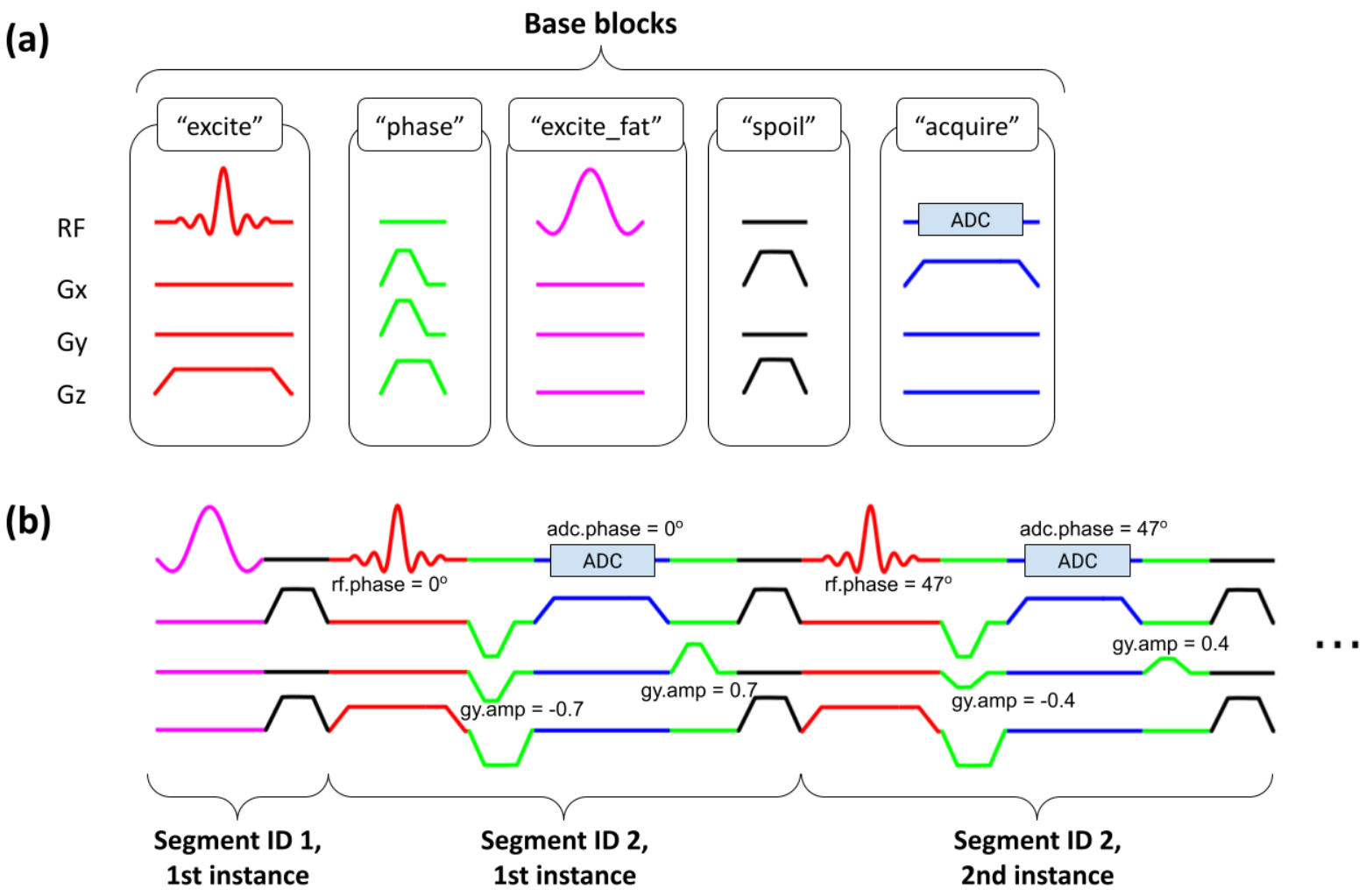


Pulseq format: brief review

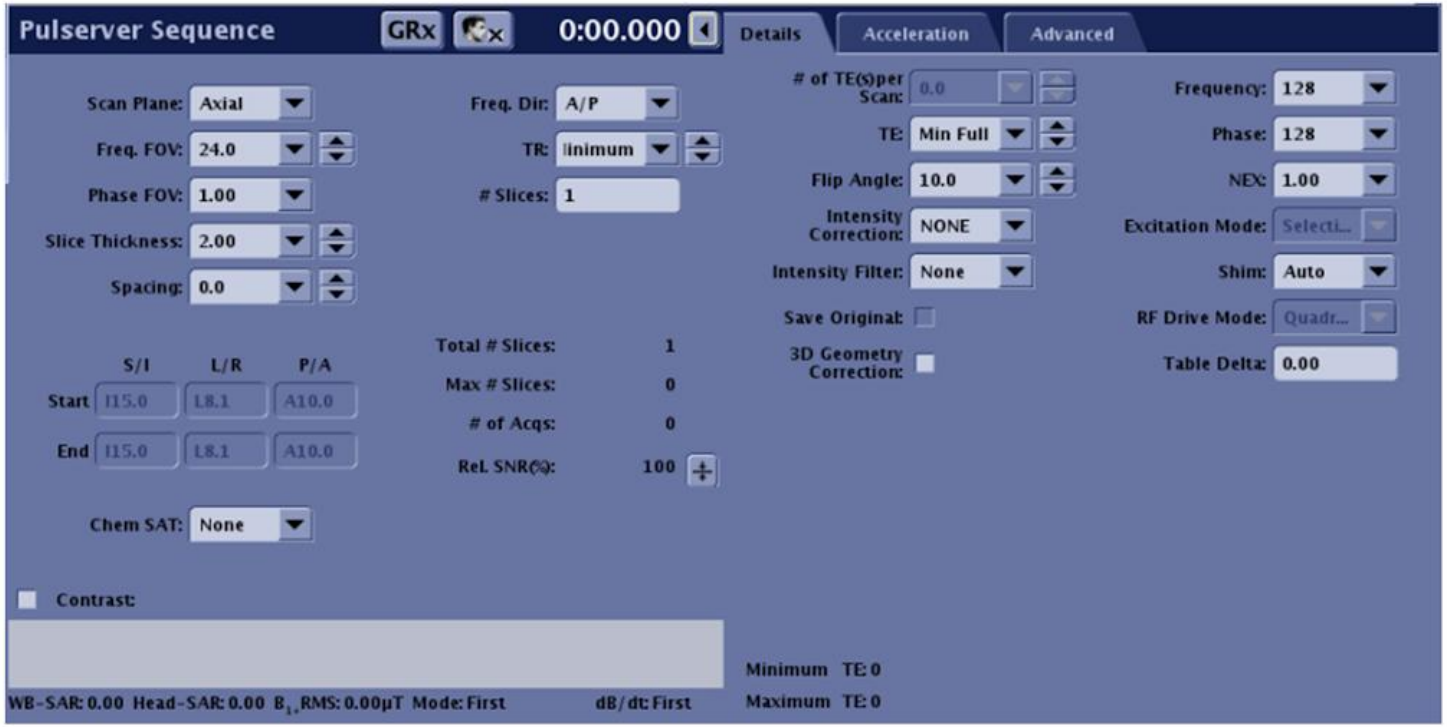


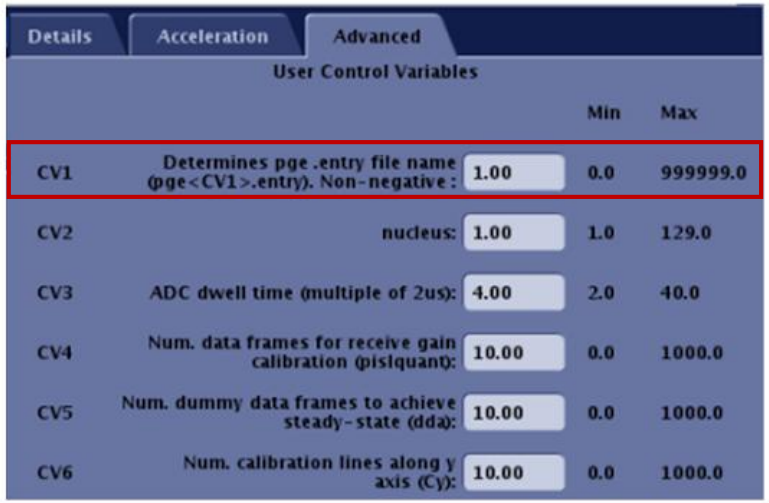
Does not enforce a structure → simple but potentially inefficient (depending on the interpreter constraints)

Efficient Pulseq builder



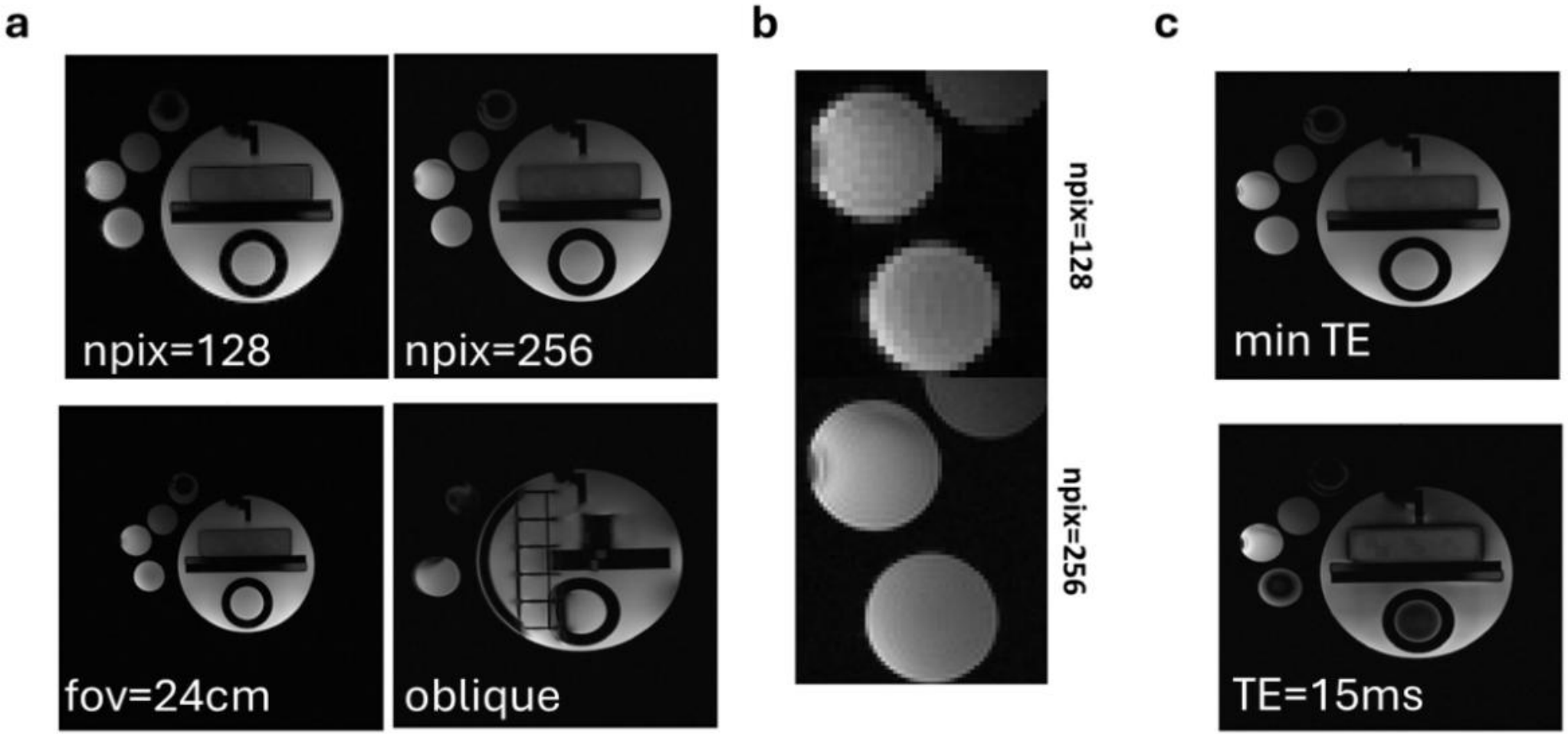
GE interpreter example: User Interface





Same interface as product sequence

Results: phantom test



Conclusion

- We achieved prescription-time design of PyPulseq sequences with similar latency as native sequences
- No external hardware was required
- Extend other interpreters
- Implement vendor-agnostic on-line reconstruction (e.g., via Gadgetron)

Thank you for your attention!